Advanced Econometrics I
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Juergen Meinecke
Research School of Economics
ANU

Problem Set 10—Suggested Answer

Setup. The normal regression model is $Y_i = X_i'\beta + e_i$ with $e_i \mid X_i \sim \mathrm{N}(0, \sigma_e^2)$, so the conditional density of Y_i given $X_i = x$ is

$$f_{Y|X}(y \mid x, \beta, \sigma_e^2) = \frac{1}{\sqrt{2\pi\sigma_e^2}} \exp\left(-\frac{(y - x'\beta)^2}{2\sigma_e^2}\right).$$

The parameter vector is $\theta = (\beta', \sigma_e^2)' \in \mathbb{R}^K \times \mathbb{R}_{++}$. Throughout, let X denote the $N \times K$ design matrix with i-th row X_i'.

(i) **Likelihood function.**

Treating the observations as conditionally independent given their regressors, the likelihood function is the product of the conditional densities evaluated at the data:

$$\begin{aligned}
L(\beta, \sigma_e^2) &= \prod_{i=1}^N f_{Y|X}(y_i \mid x_i, \beta, \sigma_e^2) \\
&= \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma_e^2}} \exp\left(-\frac{(y_i - x_i'\beta)^2}{2\sigma_e^2}\right) \\
&= \left(2\pi\sigma_e^2\right)^{-N/2} \exp\left(-\frac{1}{2\sigma_e^2} \sum_{i=1}^N (y_i - x_i'\beta)^2\right).
\end{aligned}$$

(ii) **Log likelihood function.**

Taking logarithms (a strictly increasing transformation that preserves the maximiser):

$$\begin{aligned}
\ln L(\beta, \sigma_e^2) &= -\frac{N}{2} \ln\left(2\pi\sigma_e^2\right) - \frac{1}{2\sigma_e^2} \sum_{i=1}^N (y_i - x_i'\beta)^2 \\
&= -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma_e^2) - \frac{1}{2\sigma_e^2} \sum_{i=1}^N (y_i - x_i'\beta)^2.
\end{aligned}$$

(iii) **Maximum likelihood estimators.**

We maximise $\ln L$ jointly over β and σ_e^2.

First-order condition for β.

$$\frac{\partial \ln L}{\partial \beta} = \frac{1}{\sigma_e^2} \sum_{i=1}^N x_i (y_i - x_i'\beta) = 0.$$

For any fixed $\sigma_e^2 > 0$, dividing through by $1/\sigma_e^2$ does not change the solution, so maximising over β is equivalent to minimising $\sum_{i=1}^N (y_i - x_i'\beta)^2$. That is the OLS objective. In matrix form the FOC becomes $X'(Y - X\beta) = 0$, giving

$$\hat{\beta}^{\text{ML}} = (X'X)^{-1}X'Y = \hat{\beta}^{\text{OLS}}.$$

The MLE for β is identical to the OLS estimator.

Second-order condition for β. $\partial^2 \ln L/\partial\beta\partial\beta' = -(1/\sigma_e^2)X'X$, which is negative definite (assuming $\text{rank}(X) = K$), confirming this is a maximum.

First-order condition for σ_e^2. Substituting $\hat{\beta}^{\text{ML}}$ and writing $\hat{e}_i := y_i - x_i'\hat{\beta}^{\text{ML}}$:

$$\frac{\partial \ln L}{\partial \sigma_e^2} = -\frac{N}{2\sigma_e^2} + \frac{1}{2\sigma_e^4}\sum_{i=1}^N \hat{e}_i^2 = 0.$$

Multiplying both sides by $2\sigma_e^4$:

$$-N\sigma_e^2 + \sum_{i=1}^N \hat{e}_i^2 = 0 \implies \hat{\sigma}_e^{2,\text{ML}} = \frac{\hat{e}'\hat{e}}{N}.$$

Remark: Unlike the unbiased OLS variance estimator $s^2 = \hat{e}'\hat{e}/(N-K)$, the MLE uses divisor N and is therefore biased downward (it does not account for the K estimated coefficients).

(iv) **Score and Hessian.**

Work with the log density of a single observation:

$$\ln f_{Y|X}(y \mid x, \beta, \sigma_e^2) = -\tfrac{1}{2}\ln(2\pi) - \tfrac{1}{2}\ln(\sigma_e^2) - \frac{(y - x'\beta)^2}{2\sigma_e^2}.$$

Score with respect to β ($K \times 1$ vector):

$$\frac{\partial \ln f_{Y|X}}{\partial \beta}(y \mid x, \beta, \sigma_e^2) = -\frac{1}{2\sigma_e^2}\cdot 2(y - x'\beta)(-x) = \frac{x(y - x'\beta)}{\sigma_e^2}.$$

Score with respect to σ_e^2 (scalar):

$$\frac{\partial \ln f_{Y|X}}{\partial \sigma_e^2}(y \mid x, \beta, \sigma_e^2) = -\frac{1}{2\sigma_e^2} + \frac{(y - x'\beta)^2}{2\sigma_e^4}.$$

Hence the score vector is

$$S(y \mid x, \beta, \sigma_e^2) = \begin{pmatrix} \dfrac{x(y - x'\beta)}{\sigma_e^2} \\ \dfrac{-1}{2\sigma_e^2} + \dfrac{(y - x'\beta)^2}{2\sigma_e^4} \end{pmatrix}.$$

Hessian. Differentiating the score once more:

$$H_{11} = \frac{\partial^2 \ln f_{Y|X}}{\partial\beta\partial\beta'} = \frac{\partial}{\partial\beta'}\left[\frac{x(y-x'\beta)}{\sigma_e^2}\right] = \frac{x(-x')}{\sigma_e^2} = -\frac{xx'}{\sigma_e^2} \qquad (K \times K)$$

$$H_{12} = \frac{\partial^2 \ln f_{Y|X}}{\partial\beta\,\partial\sigma_e^2} = \frac{\partial}{\partial\sigma_e^2}\left[\frac{x(y-x'\beta)}{\sigma_e^2}\right] = -\frac{x(y-x'\beta)}{\sigma_e^4} \qquad (K \times 1)$$

$$H_{21} = \frac{\partial^2 \ln f_{Y|X}}{\partial\sigma_e^2\,\partial\beta'} = \frac{\partial}{\partial\beta'}\left[-\frac{1}{2\sigma_e^2} + \frac{(y-x'\beta)^2}{2\sigma_e^4}\right] = \frac{2(y-x'\beta)(-x')}{2\sigma_e^4} = -\frac{(y-x'\beta)x'}{\sigma_e^4} \qquad (1 \times K)$$

$$H_{22} = \frac{\partial^2 \ln f_{Y|X}}{\partial(\sigma_e^2)^2} = \frac{\partial}{\partial\sigma_e^2}\left[-\frac{1}{2\sigma_e^2} + \frac{(y-x'\beta)^2}{2\sigma_e^4}\right] = \frac{1}{2\sigma_e^4} - \frac{(y-x'\beta)^2}{\sigma_e^6} \qquad \text{(scalar)}$$

Therefore

$$H(y \mid x, \beta, \sigma_e^2) = \begin{pmatrix} -\dfrac{xx'}{\sigma_e^2} & -\dfrac{x(y-x'\beta)}{\sigma_e^4} \\ -\dfrac{(y-x'\beta)x'}{\sigma_e^4} & \dfrac{1}{2\sigma_e^4} - \dfrac{(y-x'\beta)^2}{\sigma_e^6} \end{pmatrix}.$$

(v) **Fisher information matrix.**

The Fisher information is $I(\beta, \sigma_e^2) := -\mathrm{E}(H(Y_i \mid X_i, \beta, \sigma_e^2))$. We evaluate each block using two key facts that follow from $e_i|X_i \sim \mathrm{N}(0, \sigma_e^2)$:

$$\mathrm{E}(Y_i - X_i'\beta \mid X_i) = 0 \quad \text{and} \quad \mathrm{E}\big((Y_i - X_i'\beta)^2 \mid X_i\big) = \sigma_e^2.$$

Block $-\mathbf{E}(H_{11})$:

$$-\mathrm{E}\left(-\frac{X_iX_i'}{\sigma_e^2}\right) = \frac{\mathrm{E}(X_iX_i')}{\sigma_e^2}.$$

Block $-\mathbf{E}(H_{12})$: By iterated expectations,

$$-\mathrm{E}\left(-\frac{X_i(Y_i - X_i'\beta)}{\sigma_e^4}\right) = \frac{\mathrm{E}[X_i\,\mathrm{E}(Y_i - X_i'\beta \mid X_i)]}{\sigma_e^4} = \frac{\mathrm{E}(X_i \cdot 0)}{\sigma_e^4} = 0.$$

By symmetry, $-\mathrm{E}(H_{21}) = 0'$.

Block $-\mathbf{E}(H_{22})$:

$$-\mathrm{E}\left(\frac{1}{2\sigma_e^4} - \frac{(Y_i - X_i'\beta)^2}{\sigma_e^6}\right) = -\frac{1}{2\sigma_e^4} + \frac{\mathrm{E}(Y_i - X_i'\beta)^2}{\sigma_e^6} = -\frac{1}{2\sigma_e^4} + \frac{\sigma_e^2}{\sigma_e^6} = \frac{1}{2\sigma_e^4}.$$

Assembling the blocks:

$$I(\beta, \sigma_e^2) = \begin{pmatrix} \dfrac{\mathrm{E}(X_iX_i')}{\sigma_e^2} & 0 \\ 0' & \dfrac{1}{2\sigma_e^4} \end{pmatrix}.$$

Remark: The block-diagonal structure means that, asymptotically, $\hat{\beta}^{\mathrm{ML}}$ and $\hat{\sigma}_e^{2,\mathrm{ML}}$ are uncorrelated—a useful feature that is special to the normal regression model.

(vi) **Asymptotic distribution.**

By the general MLE asymptotic normality theorem,

$$\sqrt{N}\begin{pmatrix}\hat{\beta}^{\mathrm{ML}}-\beta \\ \hat{\sigma}_e^{2,\mathrm{ML}}-\sigma_e^2\end{pmatrix} \xrightarrow{\mathrm{d}} \mathrm{N}\left(\begin{pmatrix}0\\0\end{pmatrix},\ I(\beta,\sigma_e^2)^{-1}\right).$$

Since $I(\beta,\sigma_e^2)$ is block diagonal, its inverse is simply

$$I(\beta,\sigma_e^2)^{-1}=\begin{pmatrix}\sigma_e^2\,\mathrm{E}(X_iX_i')^{-1} & 0 \\ 0' & 2\sigma_e^4\end{pmatrix},$$

giving the marginal asymptotic distributions

$$\sqrt{N}\left(\hat{\beta}^{\mathrm{ML}}-\beta\right) \xrightarrow{\mathrm{d}} \mathrm{N}\big(0,\ \sigma_e^2\mathrm{E}(X_iX_i')^{-1}\big)$$

and

$$\sqrt{N}\big(\hat{\sigma}_e^{2,\mathrm{ML}}-\sigma_e^2\big) \xrightarrow{\mathrm{d}} \mathrm{N}\big(0,\ 2\sigma_e^4\big).$$

Remark on $\hat{\beta}^{ML}$: The asymptotic variance $\sigma_e^2\mathrm{E}(X_iX_i')^{-1}$ is the same as the asymptotic variance of the OLS estimator under homoskedasticity—consistent with $\hat{\beta}^{\mathrm{ML}}=\hat{\beta}^{\mathrm{OLS}}$.

Remark on $\hat{\sigma}_e^{2,ML}$: The asymptotic variance $2\sigma_e^4$ can be verified directly: since $e_i\sim\mathrm{N}(0,\sigma_e^2)$, we have $(e_i/\sigma_e)^2\sim\chi^2(1)$, and the variance of a $\chi^2(1)$ variable is 2, so $\mathrm{Var}(e_i^2)=2\sigma_e^4$.